Luis A. Ortega

Postdoctoral Researcher | Aalborg University | Copenhagen, Denmark

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Experience

Postdoctoral Researcher, *Aalborg University, Copenhagen* — 2026-Present
Postdoctoral research in Probabilistic Machine Learning and uncertainty estimation.

Teaching Assistant and Research Personnel, *Autonomous University of Madrid* — 11/2021-11/2025
Ph.D. student supervised by Daniel Hernández-Lobato. Taught Python programming fundamentals for bachelor students and Bayesian methods for master students.

Visitor Researcher, *University of Cambridge* — 09/2023-12/2023
Research on uncertainty estimation in large language models with José Miguel Hernández-Lobato.

Research Assistant, *University of Almería* — 02/2021-12/2021
Worked with Andrés R. Masegosa studying the effect of diversity on deep neural network ensembles.

Publications

Scalable Linearized Laplace Approximation via Surrogate Neural Kernel — ESANN 2026 | Spotlight talk
Luis A. Ortega, Simon Rodriguez-Santana, Daniel Hernandez-Lobato | **PDF** **Code**

Improving the Linearized Laplace Approximation via Quadratic Approximations — ESANN 2026
Pedro Jimenez, **Luis A. Ortega**, Pablo Morales-Alvarez, Daniel Hernandez-Lobato | **PDF**

A Large Deviation Theory Analysis on the Implicit Bias of SGD — Neurocomputing 2026
Luis A. Ortega, Andres R. Masegosa | **PDF** **Code**

PAC-Chernoff Bounds: Understanding Generalization in the Interpolation Regime — JAIR | ECAI 2025 Spotlight
Andres R. Masegosa, **Luis A. Ortega** | **PDF** **Code**

PAC-Bayes-Chernoff Bounds for Unbounded Losses — NeurIPS 2024
Ioar Casado, **Luis A. Ortega**, Aritz Perez, Andres R. Masegosa | **PDF**

Variational Linearized Laplace Approximation for Bayesian Deep Learning — ICML 2024
Luis A. Ortega, Simon Rodriguez-Santana, Daniel Hernandez-Lobato | **PDF** **Code**

The Cold Posterior Effect Indicates Underfitting — TMLR 2024
Yijie Zhang, Yi-Shan Wu, **Luis A. Ortega**, Andres R. Masegosa | **PDF** **Code**

Deep Variational Implicit Processes — ICLR 2023
Luis A. Ortega, Simon Rodriguez-Santana, Daniel Hernandez-Lobato | **PDF** **Code**

Diversity and Generalization in Neural Network Ensembles — AISTATS 2022
Luis A. Ortega, Rafael Cabanas, Andres R. Masegosa | **PDF** **Code**

Correcting Model Bias with Sparse Implicit Processes — ICML Workshop 2022
Simon Rodriguez-Santana, **Luis A. Ortega**, Daniel Hernandez-Lobato, Bryan Zaldivar | **PDF** **Code**

Ongoing Research

Loss-Tail SGD: Generalization-Aware Optimization via Batch Rejection Sampling — Ongoing
Filters candidate mini-batches through loss-tail rejection sampling to reduce data-dependent bias, improving generalization and calibration without changing the optimizer or objective.

Flow-Transformed Implicit Processes for Function-Space Variational Inference — Ongoing
Uses normalizing flows over finite sampled-function combinations to enrich function-space variational inference with implicit-process priors.
Preprint

Contrastive Linearized Laplace: Loss Geometry, Valid Repairs, and Reliable Cores — Ongoing
Reviews when linearized Laplace is valid for contrastive objectives and uses repaired posteriors to identify stable downstream cores.

Fixed-Mean Gaussian Processes for ad-hoc Bayesian Deep Learning — Ongoing
Converting models to Bayesian predictors by creating a Gaussian process with fixed predictive mean.
Preprint

Regularization as Estimation, A PAC-Bayes-Chernoff Approach — Ongoing
A prescriptive framework that reframes regularization as a statistical estimation problem.

Revisiting the Marginal Likelihood through a PAC-Bayesian Lens — Ongoing
Generalization in Bayesian models depends on factors beyond marginal likelihood alone.

Education

Ph.D. Student, *Autonomous University of Madrid* 11/2021-11/2025
Thesis: Uncertainty Estimation and Generalization Bounds for Modern Deep Learning

B.S. in Physics, *National Distance Education University* 2024-Present

M.S. in Data Science, *Autonomous University of Madrid* 2020-2022
Master thesis: Deep Variational Implicit Processes

B.S. in Computer Science, *University of Granada* 2015-2020

B.S. in Mathematics, *University of Granada* 2015-2020

Honors & Awards

Santander-UAM Scholarship 2023
Research stay on uncertainty estimation in large language models at the University of Cambridge.

FPI-UAM Scholarship 2021
Competitive predoctoral contract for training research personnel at the Autonomous University of Madrid.

Research Collaboration Scholarship 2020
Department of Computer Science, Autonomous University of Madrid.

Highest mark on bachelor's thesis 2020
10/10 for the thesis Statistical Models with Variational Methods at the University of Granada.

Open Source Contributions

BayesiPy
Post-hoc Bayesian deep-learning suite with VaLLA, ELLA, FMGP, SNGP, and MFVI components.

Laplace
Functional Gaussian-process Laplace implementation for the broader Laplace approximation toolkit.

apuntesDGIIM
Open notes for the double degree in computer science and mathematics at Granada.

Skills

Languages	Python, C++
ML frameworks	PyTorch, JAX, TensorFlow, Keras
Research methods	Bayesian deep learning, Variational inference, Gaussian processes, Laplace approximations, Uncertainty quantification, PAC-Bayes and generalization bounds, Deep ensembles
Research practice	Research software, Scientific writing, Collaborative research, Teaching